

Algorithmic Pseudocode Sample 28

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

```
function TRANSPORT( $\mathbf{M}, \beta, \mathbf{g}, \Delta t$ )  $\triangleright$  Input moments, natural parameters, and gauge
parameters for all cells
   $\beta \leftarrow \text{NEWTONOPTIMIZATION}(\mathbf{M}, \beta, \phi(\mathbf{u}; \mathbf{g}))$   $\triangleright$  Solve the natural parameters by Alg. ?? in
the gauge  $\mathbf{g}$ 
   $\mathbf{F} \leftarrow \text{COMPUTEFLUXES}(\beta, \phi(\mathbf{u}; \mathbf{g}))$   $\triangleright$  Compute fluxes of the statistics  $\phi(\mathbf{u}; \mathbf{g})$  by (??)
   $\mathbf{M}_{\pm 1}, \mathbf{F}_{\pm 1} \leftarrow \text{SPATIALGAUGETRANSFORMATION}(\mathbf{M}, \mathbf{F}, \mathbf{g})$   $\triangleright$  Perform gauge transformation as
in (??)
   $\mathbf{M} \leftarrow \text{FINITEVOLUMESTEP}(\mathbf{M}, \mathbf{F}, \mathbf{M}_{\pm 1}, \mathbf{F}_{\pm 1}, \mathbf{g}, \Delta t)$   $\triangleright$  One step forward in time by (??)
  return  $\mathbf{M}, \beta, \mathbf{g}$ 
end function
function COLLISION( $\mathbf{M}, \beta, \mathbf{g}, \Delta t$ )  $\triangleright$  Input moments, natural parameters, and gauge parameters
for all cells
   $\mathbf{M} \leftarrow \text{SOURCETERM}(\mathbf{M}, \mathbf{g}, \Delta t)$   $\triangleright$  Compute source term by operator splitting (??)
  return  $\mathbf{M}, \beta, \mathbf{g}$ 
end function
function STEP( $\mathbf{M}, \beta, \mathbf{g}$ )  $\triangleright$  Input moments, natural parameters, and gauge parameters for all
cells
   $\mathbf{M}, \beta, \mathbf{g} \leftarrow \text{COLLISION}(\mathbf{M}, \beta, \mathbf{g}, \Delta t/2)$   $\triangleright$  Compute collision term by operator splitting
   $\mathbf{M}, \beta, \mathbf{g} \leftarrow \text{TRANSPORT}(\mathbf{M}, \beta, \mathbf{g}, \Delta t)$   $\triangleright$  Compute Transport term by operator splitting
   $\mathbf{M}, \beta, \mathbf{g} \leftarrow \text{COLLISION}(\mathbf{M}, \beta, \mathbf{g}, \Delta t/2)$   $\triangleright$  Compute collision term by operator splitting
   $\mathbf{g}_H \leftarrow \text{COMPUTEGAUGEPARAMETERS}(\mathbf{M}, \mathbf{g})$   $\triangleright$  Compute the Hermite gauge parameters by
(??)
   $\mathbf{M}_H, \beta_H \leftarrow \text{GAUGETRANSFORMATION}(\mathbf{M}, \beta, \mathbf{g}_H, \mathbf{g})$   $\triangleright$  Transform into the Hermite gauge
using (??) and (??)
  return  $\mathbf{M}_H, \beta_H, \mathbf{g}_H$ 
end function
```
